1) a)

When $x = 0$, $y = 6$:

$6 = 2p - q\cosh 0 \Rightarrow 6 = 2p - q$ (1)

When $y = 0$, $x = 3$:

$0 = 2p - q\cosh 3$ (2)

$p = 3.33084\ldots = 3.33$ (3 SF)

$q = 0.661692\ldots = 0.662$ (3 SF)

1) b)

$y = 2(3.33084\ldots) - 0.661692\ldots\cosh x$

$y|_{x=2} = 6.6616\ldots - 0.661692\ldots\cosh 2$

$= 4.1722\ldots$

The vehicle is 4 metres tall and the tunnel is at least 4.17 metres tall (3 SF) if the vehicle drives so that its centre is also at $x = 0$. So yes, the vehicle can pass through the tunnel without touching the sides.

2)

$\frac{1}{0.75-0}\int_0^{0.75}(60 - 15\tanh 0.8t)\,\mathrm{d}t$

$= \frac{4}{3}\left[60t - \frac{75}{4}\ln|\cosh 0.8t|\right]_0^{0.75}$

$= \frac{4}{3}\left(\left(45 - \frac{75}{4}\ln\left|\cosh\frac{3}{5}\right|\right) - (0 - 0)\right)$

$= 60 - 25\ln\left|\cosh\frac{3}{5}\right|$

$= 55.7466\ldots$

$= 55.75℃$ (2 DP)

3) a)

As $t \to \infty$, $\tanh(0.12t) \to 1$, so $v \to 55 \times 1 = 55$

So the model suggests that the terminal velocity is 55 ms^{-1}

3) b)

90% of terminal velocity $= 0.9 \times 55 = 49.5$

$55\tanh(0.12t) = 49.5$

$\Rightarrow \tanh(0.12t) = 0.9$

$\Rightarrow 0.12t = \operatorname{artanh} 0.9$

$\Rightarrow t = \frac{25}{3} \times \frac{1}{2}\ln\left(\frac{1+0.9}{1-0.9}\right) = \frac{25}{6}\ln 19$

$= 12.268 \ldots$

$= 12.27$ seconds (2 DP)

4) a)

$V = \pi \int_{-4}^{4} \left(2 + \cosh\frac{x}{2}\right)^2 \mathrm{d}x$

$= \pi \int_{-4}^{4} \left(4 + 4\cosh\frac{x}{2} + \cosh^2\frac{x}{2}\right) \mathrm{d}x$

Since $\cosh^2 A = \frac{1}{2}\cosh 2A + \frac{1}{2}$

$V = \pi \int_{-4}^{4} \left(4 + 4\cosh\frac{x}{2} + \frac{1}{2}\cosh x + \frac{1}{2}\right) \mathrm{d}x$

$= \pi \int_{-4}^{4} \left(\frac{9}{2} + 4\cosh\frac{x}{2} + \frac{1}{2}\cosh x\right) \mathrm{d}x$

$= \pi \left[\frac{9}{2}x + 8\sinh\frac{x}{2} + \frac{1}{2}\sinh x\right]_{-4}^{4}$

$= \pi \left[\left(18 + 8\sinh 2 + \frac{1}{2}\sinh 4\right) - \left(-18 + 8\sinh(-2) + \frac{1}{2}\sinh(-4)\right)\right]$

$= 381.1370271 \ldots$

$\therefore V = 381$ m^3 to the nearest cubic metre.

4) b)

The pipe fills at a rate of 0.85 m^{3}s^{-1}, so $\frac{381}{0.85} = 448.235 \ldots$

So it takes approximately 448 s

5) a)

When $I = fI_0$:

$fI_0 = I_0 \operatorname{sech}(kx)$

$\Rightarrow f = \operatorname{sech}(kx)$

$\Rightarrow \frac{1}{f} = \cosh(kx)$

$\Rightarrow kx = \operatorname{arcosh}\left(\frac{1}{f}\right)$

$\Rightarrow x = \frac{1}{k}\operatorname{arcosh}\left(\frac{1}{f}\right)$

5) b)

$f = 0.4$ when $x = 12$

$12 = \frac{1}{k}\operatorname{arcosh}\left(\frac{1}{0.4}\right)$

$\Rightarrow k = \frac{1}{12}\operatorname{arcosh}\frac{5}{2}$

$= \frac{1}{12}\ln\left(\frac{5}{2} + \sqrt{\left(\frac{5}{2}\right)^2 - 1}\right)$

$= \frac{1}{12}\ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$

$= 0.1305666 \ldots$

$\therefore k = 0.131$ (3 SF)

5) c)

$I_0 = 120$

$I = 120 \operatorname{sech}(kx)$

$\frac{\mathrm{d}I}{\mathrm{d}x} = -120k \tanh(kx) \operatorname{sech}(kx)$

5) d)

$\left.\frac{\mathrm{d}I}{\mathrm{d}x}\right|_{x=8} = -120 \times 0.131 \tanh(8 \times 0.131) \operatorname{sech}(8 \times 0.131)$

$= -7.66741 \ldots$

$= -7.67$ (3 SF)

5) e)

$\frac{\mathrm{d}I}{\mathrm{d}x} = -120 \times 0.131 \tanh(0.131x)\,\mathrm{sech}(0.131x)$

$= -15.72 \tanh(0.131x)\,\mathrm{sech}(0.131x)$

For $x \geq 0$, we have $\tanh(0.131x) \geq 0$ and $\mathrm{sech}(0.131x) > 0$

So $\frac{\mathrm{d}I}{\mathrm{d}x} \leq 0$ for $x \geq 0$

As x increases, $\mathrm{sech}(0.131x)$ decreases, and $\tanh(0.131x)$ increases but is bounded from above by 1.

So $\tanh(0.131x)\,\mathrm{sech}(0.131x)$ decreases for $x \geq 0$

Therefore, $\left|\frac{\mathrm{d}I}{\mathrm{d}x}\right|$ decreases as x increases for $x \geq 0$

This suggests that the light intensity decreases most rapidly near the surface of the water, and that the rate at which intensity decreases becomes smaller with increasing depth.

5) f)

$\mathrm{sech}\, u \approx 1 - \frac{u^2}{2}$ for small u

So $\mathrm{sech}(kx) \approx 1 - \frac{(kx)^2}{2} = 1 - \frac{k^2x^2}{2}$ for small values of x

Therefore, $I = 120\,\mathrm{sech}(kx) \approx 120\left(1 - \frac{k^2x^2}{2}\right) = 120 - 60k^2x^2$

This approximation suggests that, close to the surface, the intensity of light decreases approximately quadratically with depth. This supports the model's prediction of a rapid initial decrease in intensity as the light enters the water.

6) a)

RHS: $\cosh kt + \sinh kt$

$\equiv \frac{\mathrm{e}^{kt}+\mathrm{e}^{-kt}}{2} + \frac{\mathrm{e}^{kt}-\mathrm{e}^{-kt}}{2}$

$\equiv \frac{1}{2}(\mathrm{e}^{kt} + \mathrm{e}^{kt}) + \frac{1}{2}(\mathrm{e}^{-kt} - \mathrm{e}^{-kt})$

$\equiv \mathrm{e}^{kt}$:LHS as required

6) b)

$\frac{d^2x}{dt^2} - 9x = 0$

AE: $m^2 - 9 = 0 \Rightarrow m = 3$ or $m = -3$

$\therefore x = Pe^{3t} + Qe^{-3t}$

$= P(\cosh 3t + \sinh 3t) + Q(\cosh(-3t) + \sinh(-3t))$

$= P(\cosh 3t + \sinh 3t) + Q(\cosh 3t - \sinh 3t)$

$= (P + Q)\cosh 3t + (P - Q)\sinh 3t$

$= A\cosh 3t + B\sinh 3t$ where $A = P + Q$ and $B = P - Q$

When $t = 0$, $x = 2$:

$2 = A\cosh 0 + B\sinh 0 \Rightarrow 2 = A$

So $x = 2\cosh 3t + B\sinh 3t$

$v = \frac{dx}{dt} = 6\sinh 3t + 3B\cosh 3t$

When $t = 0$, $v = 3$:

$3 = 6\sinh 0 + 3B\cosh 0 \Rightarrow 3 = 3B \Rightarrow B = 1$

$\therefore x = 2\cosh 3t + \sinh 3t$

6) c)

$R\cosh(3t + \alpha) \equiv R(\cosh 3t\cosh\alpha + \sinh 3t\sinh\alpha)$

$\equiv R\cosh\alpha\cosh 3t + R\sinh\alpha\sinh 3t$

$R\cosh\alpha = 2 \Rightarrow R = \frac{2}{\cosh\alpha}$

$R\sinh\alpha = 1 \Rightarrow R = \frac{1}{\sinh\alpha}$

$\frac{2}{\cosh\alpha} = \frac{1}{\sinh\alpha} \Rightarrow \frac{\sinh\alpha}{\cosh\alpha} = \frac{1}{2}$

$\Rightarrow \tanh\alpha = \frac{1}{2}$

$\Rightarrow \alpha = \operatorname{artanh}\frac{1}{2}$

$= \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)$

$= \frac{1}{2}\ln 3$

$\therefore R = \frac{2}{\cosh\left(\frac{1}{2}\ln 3\right)} = \frac{2}{\frac{e^{\frac{1}{2}\ln 3} + e^{-\frac{1}{2}\ln 3}}{2}} = \frac{4}{e^{\frac{1}{2}\ln 3} + e^{-\frac{1}{2}\ln 3}} = \frac{4}{e^{\ln 3^{\frac{1}{2}}} + e^{\ln 3^{-\frac{1}{2}}}}$

$= \frac{4}{\sqrt{3}+\frac{1}{\sqrt{3}}} = \frac{4\sqrt{3}}{3+1} = \sqrt{3}$

$\therefore x = \sqrt{3}\cosh\left(3t + \frac{1}{2}\ln 3\right)$

6) d)

$2\sqrt{3} = \sqrt{3}\cosh\left(3t + \frac{1}{2}\ln 3\right)$

$\Rightarrow \cosh\left(3t + \frac{1}{2}\ln 3\right) = 2$

$\Rightarrow 3t + \frac{1}{2}\ln 3 = \operatorname{arcosh} 2$

$\Rightarrow t = \frac{1}{3}\operatorname{arcosh} 2 - \frac{1}{6}\ln 3$

$= \frac{1}{3}\ln\left(2 + \sqrt{2^2 - 1}\right) - \frac{1}{6}\ln 3$

$= \frac{1}{3}\ln\left(2 + \sqrt{3}\right) - \frac{1}{6}\ln 3$

$= \frac{1}{6}\left(2\ln\left(2 + \sqrt{3}\right) - \ln 3\right)$

$= \frac{1}{6}\left(\ln\left(\left(2 + \sqrt{3}\right)^2\right) - \ln 3\right)$

$= \frac{1}{6}\left(\ln\left(7 + 4\sqrt{3}\right) - \ln 3\right)$

$= \frac{1}{6}\ln\left(\frac{7}{3} + \frac{4}{3}\sqrt{3}\right)$

7) a)

$v = A\sinh kt$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = A \times k \times \cosh kt$

$= Ak\sqrt{\cosh^2 kt}$

$= Ak\sqrt{1 + \sinh^2 kt}$

$= k\sqrt{A^2(1 + \sinh^2 kt)}$

$= k\sqrt{A^2 + A^2\sinh^2 kt}$

$= k\sqrt{(A\sinh kt)^2 + A^2}$

$= k\sqrt{v^2 + A^2}$

7) b)

$t = 0$, $a = 2$

$2 = Ak\cosh 0 \Rightarrow Ak = 2 \Rightarrow k = \frac{2}{A}$

$v = 12$, $a = 5.2$:

$5.2 = k\sqrt{12^2 + A^2}$

$\Rightarrow 5.2 = \frac{2}{A}\sqrt{144 + A^2}$

$\Rightarrow 2.6A = \sqrt{144 + A^2}$

$\Rightarrow 6.76A^2 = 144 + A^2$

$\Rightarrow 5.76A^2 = 144$

$\Rightarrow A^2 = 25$

$\Rightarrow A = 5$

7) c)

$k = \frac{2}{5} = 0.4$

7) d)

$\int_0^5 5\sinh 0.4t \, \mathrm{d}t = \left[\frac{25}{2}\cosh 0.4t\right]_0^5$

$= \left(\frac{25}{2}\cosh 2\right) - \left(\frac{25}{2}\cosh 0\right)$

$= 34.52744 \ldots$

$= 35$ metres to the nearest metre.

8) a)

The area is given by:

$$A = \int_{-6}^{6} \frac{12}{\sqrt{x^2+9}}\,\mathrm{d}x = \int_{-6}^{6} \frac{12}{\sqrt{9\left(\frac{1}{9}x^2+1\right)}}\,\mathrm{d}x = \int_{-6}^{6} \frac{12}{3\sqrt{\left(\frac{1}{3}x\right)^2+1}}\,\mathrm{d}x = 4\int_{-6}^{6} \frac{1}{\sqrt{\left(\frac{1}{3}x\right)^2+1}}\,\mathrm{d}x$$

$$= 4\left[3\,\mathrm{arsinh}\left(\frac{x}{3}\right)\right]_{-6}^{6}$$

$$= 12\ln\left(2+\sqrt{2^2+1}\right) - 12\ln\left(-2+\sqrt{(-2)^2+1}\right)$$

$$= 12\ln\left(2+\sqrt{5}\right) - 12\ln\left(-2+\sqrt{5}\right)$$

$$= 12\ln\left(\frac{2+\sqrt{5}}{-2+\sqrt{5}}\right)$$

$$= 12\ln\left(9+4\sqrt{5}\right)$$

Since $9+4\sqrt{5} = \left(2+\sqrt{5}\right)^2$ then

$$A = 12\ln\left(\left(2+\sqrt{5}\right)^2\right) = 24\ln\left(2+\sqrt{5}\right)$$

Each unit represents 2 m, so area scale factor: $2^2 = 4$

So actual area is $4 \times 24\ln\left(2+\sqrt{5}\right) = 96\ln\left(2+\sqrt{5}\right)$ m^2 s req.

8) b)

Volume of prism:

$V =$ cross-sectional area × length

$= 96\ln(2+\sqrt{5}) \times 18$

$= 1728\ln(2+\sqrt{5})$ m^3

$= 2495$ m^3

8) c)

Each hay bale occupies 1.8 m^3

So number of bales: $\frac{2495}{1.8} = 1386.111\ldots$

So 1386 hay bales.